

 *Equilux*

Pay, proven equal. No salary revealed.

The first employee-verifiable pay-transparency reporting protocol. A company proves its legally required gender pay gap report was computed from its complete payroll — sealed by each employee, vouched for by the payroll provider — while every salary stays in Midnight's private state.

Midnight Buildathon · Wave 2

Compact · TypeScript · Apache 2.0

github.com/AustinChris1/Equilux

The law now demands a number companies cannot prove.

7 June
2027

FIRST REPORTS DUE • 2026 PAY
DATA

250+

EMPLOYEES, ANNUAL • 150+
SAME DATE, TRIENNIAL

≥ 5%

UNJUSTIFIED GAP IN A
CATEGORY → JOINT PAY
ASSESSMENT

Article 9 requires seven indicators. Article 7 gives every employee the right to ask what their category earns. And today every one of those reports is **self-declared**: nobody can verify the numbers without seeing everyone's salary.

Trust me, or prove it.

Today

HR exports payroll to a spreadsheet, computes a gap, publishes a PDF. An employee cannot check they were counted. A works council cannot check nobody was dropped. A regulator can only audit by subpoenaing the whole payroll — the data the law is trying to protect.

With Equilux

Every employee seals their record. The payroll provider — not the employer — vouches for it. A zero-knowledge circuit verifies the gaps over the sealed set and publishes only the figures plus a proof that they came from all of it. Verification takes seconds. No salary ever leaves the company.

Equilux does not replace the HRIS. Payroll stays in Personio, DATEV or Workday. Equilux is the **notary on top**: it proves the set was complete and the published figure matches it.

Payroll-provider attestation. A plugin, not an app. Both built.

	WAVE 1	WAVE 2
Who vouches for records	the employer, for its own data	the payroll provider — the employer cannot
Leaving someone out	blocked only if they enrolled	blocked outright — roster declared before enrollment
Statistics proven	mean gap	mean · median · per worker category
Article 7 category pay	designed	published at k = 3 , small groups suppressed
The public site	a replica of the rules	the compiled contract, in the browser
HRIS integration	—	Personio / DATEV CSV import + party API
Circuits · tests	4 · 21	5 · 41

Trust neither side. **Verify both.**

Completeness

The payroll provider declares the headcount before anyone enrolls; the report must cover exactly that many sealed records. Leaving someone out — even by never enrolling them — is rejected.

No invented records

Every record carries an employee-held secret and a payroll-provider attestation. The circuit re-derives each commitment; a forged row fails.

No small-group leaks

A category's pay is published only when both groups have at least three people. Smaller groups show headcounts only.

Exact figures

Mean gap, median gap and every category's pay and gap are verified in-circuit — no division, no sort. One basis point off fails.

Plus an **inclusion receipt** for every employee: a Merkle path proving their record is inside the numbers, revealing nothing about anyone.

The provider vouches. The employee seals. The circuit checks everything.

01 ·
declareRoster

Roster

The payroll provider commits the headcount before enrollment opens.

02 · enroll

Seal

Employee seals salary, gender marker and category as a commitment and a nullifier.

03 · attest

Vouch

The payroll provider attests each record it recognises. The employer cannot.

04 ·
publishReport

Prove

Full roster, attested, distinct; mean, median and category gaps verified.

05 · checkReceipt

Verify

An employee proves their record is in the tree against the on-chain root.

Private state holds the payroll. Public state holds the proof.

Private state • witnesses

- salary, gender marker, worker category
- employee enrollment secret
- employer and payroll-provider secret keys
- the full 16-slot payroll witness
- every claimed figure (verified, never trusted)

Public ledger

- commitments — a depth-5 Merkle tree
- nullifier set, attested set, enrolled counter
- employer + provider keys (sealed), declared headcount
- `PayReport` : headcounts, mean + median gaps, 4 category rows ($k = 3$)
- the zero-knowledge proof for every transition

Every witness value that reaches the ledger passes an explicit `disclose()`. The compiler refuses anything else.

Three parties. Four cheats. All rejected. One proven report.

5

CIRCUITS · PROVER + VERIFIER
KEYS

41

TESTS · COMPILED CIRCUITS
VIA COMPACT-RUNTIME

0

SALARIES ON CHAIN · EVER

deployed at **518587d2...2821** · employer + payroll-provider keys (block 55)
enroll before roster → **rejected**: roster not yet declared by the payroll provider
provider declares roster of 8 (block 59)
8 enrollments · 9th → **rejected**: roster is full (blocks 63–91)
8 provider attestations (blocks 95–123)
omit · fake mean · fake median · hidden category gap → **all 4 rejected by the circuit**
publishReport **proven & finalized** · Engineering 13.88% flagged (block 140)

No Docker. No install. The real circuits, on the public URL.

EMPLOYER

PAYROLL PROVIDER

EMPLOYEE

REGULATOR

5 · WHAT THE REPORT SHOWS – AND NOTHING ELSE

MEAN GAP
9.69%
favors men

MEDIAN GAP
1.81%
favors men

HEADCOUNT
14
7 women · 7 men

WORKER CATEGORY	W / M	MEAN PAY W · M	GAP
Engineering	3 / 3	€62,000 · €72,000	13.88% ≥ 5% · assess
Sales	3 / 3	€52,000 · €54,000	3.70% < 5%
Operations	1 / 1	suppressed – fewer than 3 in a group, so no one's pay can be inferred	

A category at ≥ 5% triggers an Article 10 joint pay assessment only if the gap is also unjustified by objective, gender-neutral criteria – a human judgement the circuit does not make. Salaries on chain: 0.

Honest about what a proof can prove.

IT DOES NOT CLAIM	WHAT IS TRUE INSTEAD
Salaries hidden from HR	HR still holds payroll. Hidden from the public, colleagues, the regulator and Equilux — and the employer can no longer vouch for its own data.
An honest roster by itself	The provider declares the headcount; a colluding provider could under-declare. A works-council co-signature is next.
Judging what is justified	Category gaps $\geq 5\%$ are proven and flagged. Whether a gap is justified is a human judgement.
The full Article 9 return	Three of seven indicators: mean, median, category gap. Variable pay and quartiles follow.
Production scale · public testnet	16 records, 4 categories. Local Midnight network — preprod waits on a wallet-SDK fix.
A model of identity	Binary gender markers, as the Directive's reporting annex is written today.

Built in the open, wave by wave.

Wave 1 · delivered

Prove it works

Contract with private state · proven end to end on a Midnight network · employer / employee / regulator workspace · honest scope.

Wave 2 · this wave

The product

Payroll provider as a third party · roster · median and category gaps · $k = 3$ category pay · the compiled contract in the browser · Personio / DATEV import.

Wave 3

Fundable

Public testnet + Lace writes · variable pay and quartiles (all of Article 9) · filing pack with proof hash + verify URL · works-council co-signature · one named pilot.

The notary for Directive 2023/970.

Buyers

Works councils that don't trust the numbers. GDPR-sensitive employers. Regulators who want completeness without holding payroll.

A plugin, not a rival

Workday, Personio, DATEV compute pay. Equilux imports their export, the provider attests, and the report is proven — next to them, not instead of them.

Why Midnight

A forced legal deadline plus a problem only dual-ledger selective disclosure solves honestly. Incumbents can copy a dashboard; they cannot copy the proof.

